

LERWICK, United Kingdom

WMO: 030050

Lat: 60.139N

Lon: 1.185W

Elev: 276

StdP: 14.55

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	27.9	29.9	20.2	15.3	31.8	22.3	17.0	33.2	48.5	42.3	45.6	40.6	13.5	350	0.901

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	61.4	58.2	59.6	56.8	58.2	55.7	59.2	60.7	57.7	58.9	56.4	57.6	12.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	73.6	59.9	57.0	69.9	58.5	55.7	66.8	57.0	26.0	60.8	25.0	59.0	24.2	57.7	64.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
42.9	38.3	33.8	25.0	65.4	2.6	3.3	23.2	67.8	21.7	69.8	20.2	71.6	18.4	74.0
			23.1	60.7	2.4	2.4	21.4	62.4	20.0	63.8	18.7	65.2	17.0	66.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	39.6	38.9	40.0	42.6	46.0	50.1	53.7	54.4	51.8	47.5	43.2	40.4
	DBStd	6.66	3.86	4.36	4.09	3.68	3.56	2.99	2.91	2.68	3.16	3.90	4.47	4.89
	HDD50	1955	324	311	311	223	130	35	3	2	19	92	206	299
	HDD65	7043	789	731	776	673	590	448	350	328	396	543	655	764
	CDD50	388	0	0	0	0	6	37	118	139	73	14	1	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	17.0	20.6	20.2	19.0	16.4	14.5	14.4	13.0	13.2	15.9	18.3	19.2	19.4
Precipitation	PrecAvg	48.3	5.5	4.3	4.3	2.8	2.3	2.4	2.5	3.3	4.4	5.2	5.7	5.7
	PrecMax	61.1	11.9	9.8	8.1	5.7	5.2	5.3	6.1	7.2	7.1	9.8	9.3	10.8
	PrecMin	33.4	1.6	0.4	0.9	0.1	0.3	0.4	1.0	0.9	1.5	2.2	1.9	2.4
	PrecStd	6.4	2.0	1.9	1.6	1.2	1.1	1.0	1.1	1.4	1.4	1.7	1.7	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	48.7	49.3	52.6	58.3	62.4	65.2	64.9	60.4	56.1	52.7	51.1
		MCWB	47.0	46.9	46.2	48.0	53.4	56.5	60.4	60.3	56.6	54.0	50.9	49.4
	2%	DB	47.4	47.5	47.6	50.2	54.2	57.4	61.4	61.3	58.2	54.2	51.1	49.4
		MCWB	45.9	46.2	45.4	47.0	50.2	53.3	58.6	58.2	55.5	52.3	49.7	48.1
	5%	DB	46.3	46.3	46.6	48.6	52.4	55.6	59.5	59.7	57.0	53.1	50.2	48.3
		MCWB	44.8	44.8	44.7	46.0	49.1	52.1	57.2	57.0	54.9	51.4	48.9	47.1
	10%	DB	45.2	44.9	45.5	47.4	50.9	54.1	58.0	58.3	56.0	52.3	49.2	47.1
		MCWB	43.7	43.3	43.6	45.2	48.0	51.4	55.7	55.9	54.2	50.7	47.8	45.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.0	48.1	47.6	49.5	54.1	57.1	61.7	61.0	57.8	54.7	51.6	50.1
		MCDB	48.3	48.3	48.4	50.7	56.9	61.0	64.2	63.2	59.0	55.5	52.2	50.7
	2%	WB	46.4	46.7	46.3	47.8	51.3	54.4	59.4	59.2	56.5	53.1	50.4	48.7
		MCDB	47.1	47.3	47.1	49.1	53.4	56.4	60.8	60.7	57.3	53.7	50.9	49.3
	5%	WB	45.2	45.2	45.2	46.7	49.9	53.1	57.9	57.9	55.5	52.2	49.3	47.4
		MCDB	46.1	46.1	46.2	48.1	51.6	54.6	59.2	59.1	56.4	52.8	50.0	48.2
	10%	WB	44.0	43.7	44.0	45.7	48.7	52.2	56.4	56.6	54.7	51.3	48.1	45.9
		MCDB	45.1	44.9	45.2	47.1	50.3	53.5	57.5	57.7	55.6	52.1	49.1	47.0
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.1	6.6	7.0	7.7	7.0	6.6	6.6	6.2	6.0	5.8	6.1
		MCDBR	6.3	6.4	6.8	7.5	8.8	8.3	8.2	7.8	6.9	5.9	6.2	6.5
		MCWBR	6.9	6.5	6.1	5.5	6.0	5.6	5.6	5.4	5.2	5.6	6.9	7.5
	5% WB	MCDBR	6.4	6.1	6.6	7.0	8.0	7.4	7.6	7.2	6.1	5.5	6.1	6.6
		MCWBR	7.1	6.7	6.5	5.6	5.9	5.5	5.6	5.7	5.3	5.9	7.0	7.7
Clear-Sky Solar Irradiance	taub	0.332	0.351	0.365	0.380	0.370	0.364	0.378	0.365	0.361	0.345	0.326	0.423	
	taud	2.087	2.212	2.286	2.307	2.404	2.459	2.434	2.472	2.446	2.329	2.143	2.458	
	Ebn at Noon	124	198	239	259	272	275	268	263	242	200	126	85	
	Edh at Noon	15	23	29	34	33	32	32	29	25	20	14	10	
All-Sky Solar Radiation	RadAvg	76	243	595	1093	1526	1561	1419	1155	704	351	111	48	
	RadStd	7	28	41	72	96	157	123	108	61	34	10	3	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.61	+0.88	+1.55	N/A	N/A	N/A	-187	-223	N/A	N/A	
(47)	Regional (18 neighbors)	+0.58	N/A	+1.48	N/A	N/A	N/A	-160	-205	N/A	N/A	

Nomenclature: See separate page